

Memory Residuals: Native Long-Term Memory for Conversational Language Models via Two-Stage Slot Competition and Depth-Wise Attention Routing

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Abstract

We present *Memory Residuals*, a recurrent latent-memory primitive that augments a pretrained Transformer backbone with a fixed-size matrix $M_c \in \mathbb{R}^{K \times d}$, updated once per session through a two-stage QKV competition and read back into the model through a depth-wise attention-residual router rather than the token sequence. The design has three load-bearing properties: a parity-preserving initialisation that produces bit-identical step-0 logits to the bare backbone; a slot-axis softmax in the writer that breaks the permutation-symmetric uniform fixed point; and an off-sequence read path that lets every layer choose, per token, between current-session computation and cross-session memory. We then show that getting positive end-to-end memory benefit out of this primitive is overwhelmingly an evaluation- and training-recipe problem, not just an architectural one. Across a 24-version empirical campaign on Qwen3-0.6B and Qwen3-1.7B we catalogue six concrete failure modes (token-averaged dilution, joint-training invisibility, category-cue leak, window leakage, redaction uncertainty, and writer-objective failure) and the diagnostics that distinguish honest memory benefit from each. The combination of a frozen backbone, the readout-probe auxiliary objective, an α_{mem} load-balance floor, and training on real callback-supervised conversational data (LongMemEval) yields $\Delta_{\text{nm-m}}^{\text{cb}} = +0.34$ nats of chain-specific callback-CE reduction on LongMemEval validation – the first non-trivial, chain-specific, leak-free end-to-end memory benefit in the project – with a near-zero shuffle baseline ($\Delta_{\text{sh-m}}^{\text{cb}} = -0.003$) ruling out a generic “memory adds context” confound. The same recipe is unstable on synthetic random-code corpora and on LongMemEval+MSC merges where most chains lack callback supervision, which we read as a corpus-density and writer-supervision result rather than an architecture-level result. We release the architecture, the training recipe, the evaluation tooling, and the run ledger.

1 Introduction

A deployed conversational agent that needs to remember a user across sessions has three options today, all unsatisfactory. **Long-context concatenation** pays $O(N^2)$ in conversation length and degrades on lost-in-the-middle benchmarks past a few thousand tokens [12]. **Retrieval-augmented generation** [10] externalises memory to a vector store: it is non-differentiable, brittle when callbacks depend on implicit state rather than lexical overlap, and weak on the abstractive-recall regime where the relevant history cannot be served verbatim. **Learned recurrent memory** crosses session boundaries with a bounded-cost differentiable state, but every published incarnation [2, 8, 19] has reported some flavour of channel collapse, training

instability, or shortcut learning where the recurrent channel encodes only style cues rather than episodic content.

This paper studies the third option deliberately. We start from a position paper [11] that describes a particular recipe – a two-stage QKV-competition memory update plus an off-sequence depth-wise read via the recently-proposed Block Attention-Residual router [5] – and ask the empirical question of whether the recipe, run honestly under the recurrent training regime it is designed for, produces a chain-specific memory that survives the diagnostic battery we know is necessary in this regime.

The answer turns out to be “yes, but only on real-content corpora with callback annotations on most chains, with a frozen backbone, with the readout-probe auxiliary objective at the right weight, and only after the standard chain evaluator is replaced with one that scores callback tokens.” Each of those qualifications is empirical: we found six distinct failure modes in the wider design space, each of which can make a memory architecture look effective when it is not, or hide a real localised effect when it is. Section 5 catalogues them.

Headline result. On the LongMemEval [20] validation chains, our final recipe (**v24a**: frozen Qwen3-0.6B, $K = 128$ slots, extraction depth $L_E = 4$, slot-attention writer, depth=4 readout, readout-probe weight 0.5, α_{mem} floor weight 0.5 targeting 0.10, 1000 steps on 11.2 M tokens of LongMemEval train) reduces callback cross-entropy from $\text{CE}_{\text{nomem}} = 3.28$ nats to $\text{CE}_{\text{mem}} = 2.94$ nats, a chain-specific improvement $\Delta_{\text{nm-m}}^{\text{cb}} = +0.3445$ that the shuffle control places at $\Delta_{\text{sh-m}}^{\text{cb}} = -0.003$. This is roughly $86\times$ the effect size of the same recipe trained on a templated synthetic corpus (synthd5: +0.024), and is the first end-to-end result in the project for which the standard battery – frozen backbone, shuffle control, callback-aware metric, no template prior in the answer distribution – does not flip the conclusion.

Contributions.

1. **Architecture.** A complete reference implementation of Memory Residuals, with three load-bearing primitives: a parity-preserving routing init, a slot-axis softmax in the Stage-2 judge that breaks the symmetric uniform fixed point, and a per-position off-sequence readout that registers as a foundational source in a depth-wise router (§2).
2. **Training recipe.** A frozen-backbone, readout-probe augmented training loop with an α_{mem} load-balance floor and multi-negative InfoNCE that empirically delivers chain-specific recall in the regime where naive LM-NLL training collapses to a content-blind writer (§3).
3. **Honest-evaluation methodology.** A callback-aware standalone evaluator, an evidence-redacted memory floor, an empirical template-prior audit, a window-leakage counter, and a redaction-spread audit. We give the diagnostic that flags each of the six failure modes we encountered (§4, §5).
4. **Empirical headline.** The first non-trivial, chain-specific, leak-free end-to-end memory benefit in our 24-version campaign: $\Delta_{\text{nm-m}}^{\text{cb}} = +0.3445$ on LongMemEval validation with a near-zero shuffle baseline (§6).
5. **Negative results.** Multi-seed instability on synthetic random-code corpora, end-to-end collapse on LME+MSC merges where 93% of training chains lack callback supervision, and the joint-training Δ_{evidence} collapse on the D4v2 closed-set persona corpus – each of which we read as evidence that what the writer *sees during training* is more decisive for the recipe than the backbone scale or the precise architectural variant (§7).

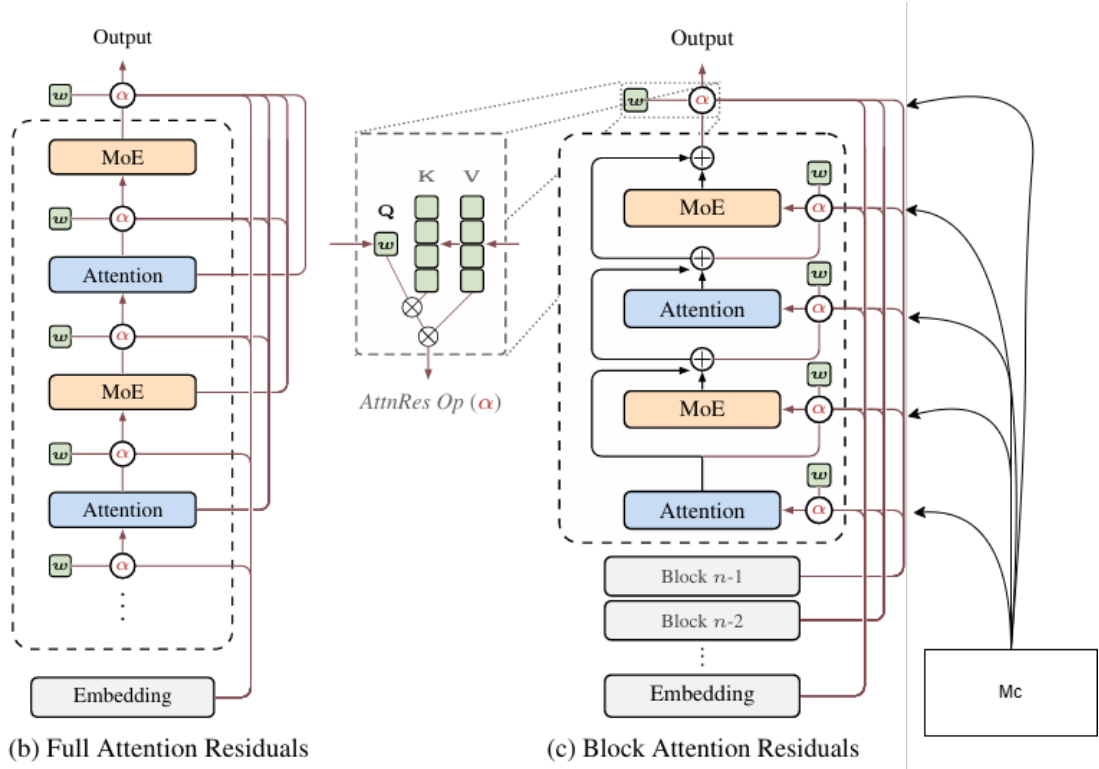


Figure 1: **From depth-wise attention routing to persistent memory.** Block Attention Residuals [5] let each layer dynamically route information from prior layer outputs instead of a uniform residual sum. Memory Residuals extend this routing pool by registering a per-position cross-session readout m^t as the foundational source v_0 , so that each layer can decide whether to use current-session computation or cross-session memory, unifying local reasoning and historical recall under one attention mechanism.

2 The Memory Residuals Architecture

The architecture augments a pretrained causal language model with a single recurrent state $M_c \in \mathbb{R}^{K \times d}$, where K is a small fixed number of slots ($K=128$ in all experiments) and d matches the backbone hidden size. The state is updated at session boundaries, never per token. There are three stages: *extract* from the just-completed session $C_t \in \mathbb{R}^{N_t \times d}$, a *judge/write* step that updates M_c , and a *read* step that injects the per-position readout m^t into the next session’s residual stream through a depth-wise router.

2.1 Stage 1 – Extract: a fixed-bandwidth latent

A learnable bank of extraction queries $M_{\text{in}} \in \mathbb{R}^{K \times d}$ cross-attends over the raw session C_t to produce a K -slot candidate state:

$$M_{\text{new}}^{(0)} = \text{Softmax} \left(\frac{M_{\text{in}} W_Q^{\text{ext}} (C_t W_K^{\text{ext}})^{\top}}{\sqrt{d}} \right) C_t W_V^{\text{ext}}. \quad (1)$$

Equation 1 is the single-layer baseline ($L_E=0$). In practice deciding what to extract from hundreds of conversation tokens is a multi-hop task: an early “my sister” may only become salient after composing it with later turns. We identify **extraction depth** L_E as a primary axis and replace (1) with a Perceiver-style refinement stack [9]:

$$M_{\text{new}}^{(\ell)} = M_{\text{new}}^{(\ell-1)} + \text{CrossAttn}^{(\ell)} \left(M_{\text{new}}^{(\ell-1)}, C_t \right), \quad \ell = 1, \dots, L_E, \quad (2)$$

with $M_{\text{new}} := M_{\text{new}}^{(L_E)}$. All intermediate states live in $\mathbb{R}^{K \times d}$, so the residual is well-typed and $L_E=0$ trivially recovers (1). We use $L_E = 4$ in all reported runs. An RMSNorm on C_t before the cross-attention (**extract_input_norm**) was added after gradient audits showed unnormalised hidden states drove extract-key gradients $\sim 50\times$ larger than expected, causing $\|M_{\text{new}}\|_F$ explosions that saturated downstream gates within 50 steps.

2.2 Stage 2 – Judge/write: slot-axis competition

The candidate M_{new} competes with the previous state through a **slot-attention** writer [13]. We concatenate old and new content into a $2K$ -row pool $P = [M_c^{t-1} \parallel M_{\text{new}}] \in \mathbb{R}^{2K \times d}$ and attend with persistent learnable judging queries $M_{\text{judge}} \in \mathbb{R}^{K \times d}$:

$$A = \text{Softmax}_{k \in [K]} \left(\frac{Q^j (K^j)^\top}{\sqrt{d}} \right) \in \mathbb{R}^{K \times 2K}, \quad M_c^t[k] = \frac{\sum_{i=1}^{2K} A[k, i] (PW_V^j)[i]}{\sum_{i=1}^{2K} A[k, i] + \varepsilon}, \quad (3)$$

where $Q^j = \text{RMSNorm}(M_{\text{judge}} W_Q^j)$ and $K^j = \text{RMSNorm}(\hat{P} W_K^j)$ are post-projection RMS-normalised (**QK-LayerNorm**; Zhai et al. 21), and $M_{\text{judge}}, \hat{P}$ are RMSNormed slot and input streams. The recurrent update is $M_c^t = \text{LayerNorm}(M_c^{t-1} + \Delta_{M_c}^t)$ through a per-slot GRU cell, so the update is recurrent rather than overwriting.

The choice of *which axis* the softmax is taken over is load-bearing. The naive Stage-2 judge takes the softmax over the $2K$ input rows: each judging query independently averages over the same input pool, so any permutation of the K rows of M_{judge} leaves the output invariant. The model is free to settle at a symmetric uniform fixed point in which all K slots hold the same averaged content, and the “zero-sum” framing is vacuous because the competition is between inputs, not between slots. In the slot-axis form of (3), two slots with near-identical query projections share each input row’s total unit mass, so any drift of one slot’s projection *must* be compensated by the other losing mass on the same inputs. The symmetric uniform configuration is no longer a fixed point of the gradient update.

Why QK-LayerNorm. Even with the slot-axis softmax, the judge is a single attention layer whose logits decay exponentially with the spectral norm of W_Q^j, W_K^j [21]. Under small-step gradient descent on the LM objective, the spectral norms drift towards small values and the per-input-row distribution drifts back towards uniform across slots. Post-projection RMSNorm decouples the magnitude of attention logits from the spectral norm of the projections, so the slot-axis softmax retains the freedom to sharpen without the LM objective having to tune two projections whose job is semantically independent of logit scale. Empirically, omitting QK-LayerNorm is the difference between a writer that specialises ($\|m^t\|/\|\text{embed}\| = 3.87$, pair/self = 0.005) and a writer that emits zero ($\|m^t\| = 0$, pair/self uninformative).

2.3 Stage 3 – Read: per-position cross-attention into M_c

When scoring the next session, every position $X \in \mathbb{R}^{S \times d}$ attends to M_c once to produce a per-position readout:

$$m^t = \text{Softmax} \left(\frac{X W_Q^r (M_c W_K^r)^\top}{\sqrt{d}} \right) M_c W_V^r \in \mathbb{R}^{S \times d}. \quad (4)$$

Each token attends over the K slots independently, so the readout is inherently query-dependent: tokens about a family conflict route mass onto slots encoding family context, while tokens about dinner plans attend elsewhere. Optionally, a depth- D_R refinement stack repeats (4) with residual updates ($D_R = 4$ in the recipe).

2.4 Off-sequence depth-wise injection

Because m^t has the same shape $S \times d$ as any sublayer output, it slots into a Block Attention Residual router [5] as a foundational source. Let a model have L layers and $v_1, \dots, v_{L-1}$ be the per-layer source tensors; we register

$$v_0 := m^t \in \mathbb{R}^{S \times d}, \quad (5)$$

and the input to any downstream layer ℓ is computed as a depth-wise softmax mix over all preceding sources:

$$h_\ell = \sum_{i=0}^{\ell-1} \alpha_{i \rightarrow \ell} v_i, \quad \alpha_{i \rightarrow \ell} = \frac{\exp(w_\ell^\top \text{RMSNorm}(v_i))}{\sum_{j=0}^{\ell-1} \exp(w_\ell^\top \text{RMSNorm}(v_j))}, \quad (6)$$

where $w_\ell \in \mathbb{R}^d$ is a learned pseudo-query for layer ℓ . We track $\alpha_{\text{mem}}^{(\ell)} := \alpha_{0 \rightarrow \ell}$ as the per-layer *memory routing mass*; the chain-aggregate α_{mem}^- is one of the principal training-time diagnostics. By integrating memory into the layer-to-layer residual stream, the model gains the ability to route memory recall *implicitly*: during a content-rich turn the recent hidden states naturally dominate the softmax, suppressing $\alpha_{\text{mem}}^{(\ell)}$; during a sparse callback turn (e.g. a bare pronoun question), the absence of dominant local features lets v_0 attract higher mass.

2.5 Parity-preserving initialisation

The router carries two scalar bias parameters `mem_bias` and `recent_bias` that initialise the softmax in (6) to one-hot on the most-recent non-memory source. With `mem_bias` = −32 the augmented model is bit-identical (in bf16) to the bare backbone at step 0; with `mem_bias` = −4 the deviation is $\sim 10^{-5}$ logit but the softmax gradient on the per-source pseudo-queries is non-saturated from step 1. We use the soft variant (± 4 bias) throughout. This matters because step-0 logit perturbations of more than a few nats make the optimiser spend most of its budget reabsorbing the perturbation rather than learning the memory channel; on Qwen3-0.6B a uniform-softmax delta-pool init (± 0) perturbs by up to ~ 34 nats and does not surpass the bare backbone on the shuffle-specificity test at matched compute.

3 Training Recipe

The training loop is recurrent truncated backpropagation through time over chain windows of k sessions. The per-step LM next-token NLL is summed over all sessions in the window so that every M_c^t receives at least one downstream gradient even at short TBPTT depth. What we describe below are the auxiliary objectives and architectural guards that make the LM-NLL gradient actually train a chain-specific memory rather than a content-blind one.

3.1 The content-blind writer collapse and how to avoid it

Across early versions (v3–v15) we repeatedly observed that LM-NLL alone, even with the slot-axis judge of Eq. 3, drifts the writer to a content-blind regime: M_c becomes useful as a generic “memory is on” prior shifting the LM toward the marginal of the callback tokens, while carrying little chain-specific binding. The mechanism is straightforward: LM-NLL is permutation-symmetric over slots and provides no penalty for uniform M_c as long as uniform M_c produces a useful m^t for next-token prediction. Once the writer is content-blind, the readout has no chain-specific signal to decode, the depth router cannot find an opening, and the system ratchets shut.

We pair three guards to prevent this dynamic.

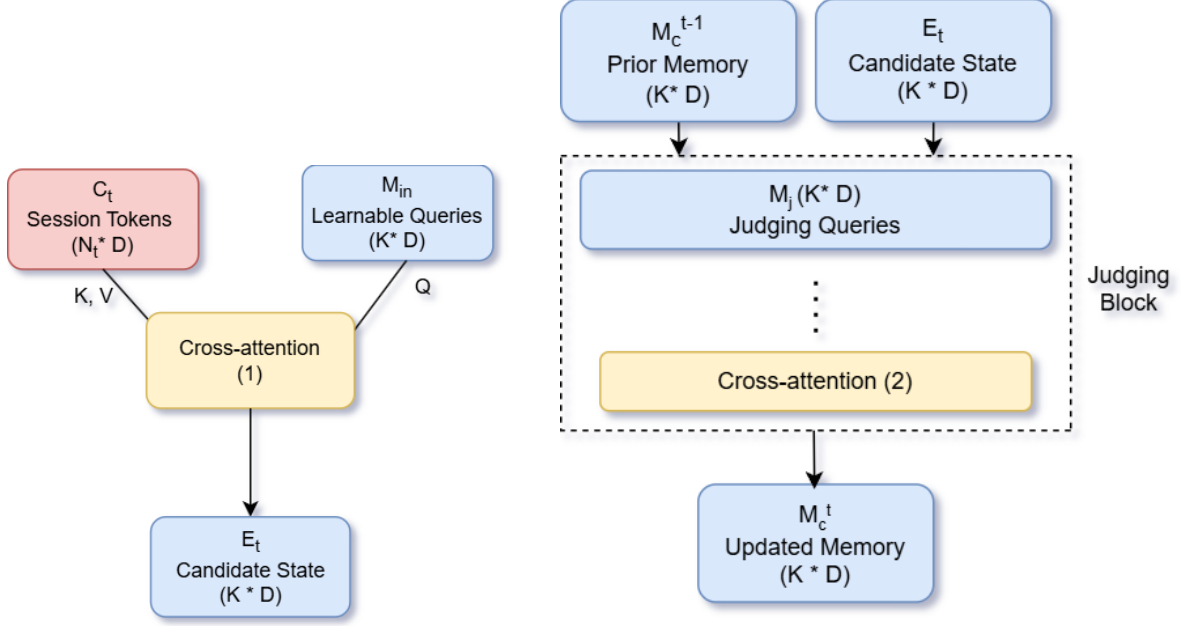


Figure 2: **The two-stage memory update.** *Left:* the extraction stack compresses the finished session C_t into a fixed-width candidate state M_{new} using learnable extraction queries M_{in} (Eqs. 1–2). *Right:* the slot-attention judge compares the prior memory M_c^{t-1} against the candidate M_{new} using persistent slot queries M_{judge} on a $2K$ -row pool, with the softmax taken over the *slot* axis (Eq. 3). Specialists, not candidates, compete.

Slot-attention symmetry break. The slot-axis softmax in Eq. 3, combined with orthogonal initialisation of the slot queries (`queries_init=orthogonal`) and additive slot positional embeddings (`slot_positional`), removes the permutation-invariant uniform fixed point. Empirically, the v13+ stack produces a writer in which the cross-chain pair/self M_c similarity stabilises at 0.004–0.075 rather than collapsing to the ~ 1.0 that the uniform fixed point would imply.

Readout-probe auxiliary objective (F3). The most decisive lever in the campaign was the **ReadoutProbeHead**: a lightweight head that consumes m^t at the callback position and predicts the answer tokens directly, with its own next-token NLL added to the total loss with weight λ_{rprobe} . Crucially, the probe consumes the actual MemoryReadout output (Eq. 4), so its gradient flows through the readout’s W_Q^r, W_K^r, W_V^r and back through M_c into the writer. This provides the writer with a direct, chain-specific extractive signal that the LM head’s long-distance gradient cannot supply.

The probe is discarded at evaluation time – it is a training scaffold, not an inference path. If the final LM-head callback evaluation improves after the probe is removed, the result means the writer has learned to store content that the normal readout can use. We swept $\lambda_{rprobe} \in \{0.3, 0.4, 0.5, 0.6, 1.0\}$ and found the end-to-end optimum at 0.5 (v21c/v24a) with substantial run-to-run noise; weights ≥ 1.0 over-pull the readout into chain-fingerprinting that does not transfer through the LM head.

α_{mem} load-balance floor. A second auxiliary loss penalises the depth-router’s memory mass α_{mem} for falling below a target τ :

$$\mathcal{L}_{floor} = \lambda_{floor} \cdot \text{ReLU}(\tau - \alpha_{mem}^-)^2. \quad (7)$$

This keeps the depth router open enough that the LM head can actually consume m^t , breaking the chicken-and-egg deadlock in which the router closes early because the writer is uninformative

Algorithm 1 One training step (single micro-batch). Backbone frozen; gradients flow only through memres parameters.

Require: chain $\{C_1, \dots, C_k\}$, current $M_c^0 = 0$

```

1: for  $t = 1, \dots, k$  do
2:   Forward  $C_t$  through backbone with depth-routed  $m^t$  from  $M_c^{t-1}$  (Eq. 4)
3:    $L_t^{\text{LM}} \leftarrow \text{NLL}(C_t)$ 
4:   if  $C_t$  is a callback session and has annotated answer mask  $A_t$  then
5:      $L_t^{\text{rprobe}} \leftarrow \text{NLL}(\text{ReadoutProbe}(m^t[A_t]), A_t)$ 
6:   end if
7:    $M_{\text{new}} \leftarrow \text{Extract}(C_t)$  ▷ Eqs. 1–2
8:    $M_c^t \leftarrow \text{LayerNorm}(M_c^{t-1} + \text{Judge}(M_c^{t-1}, M_{\text{new}}))$  ▷ Eq. 3
9: end for
10:  $\bar{L} \leftarrow \frac{1}{k} \sum_t L_t^{\text{LM}} + \lambda_{\text{rprobe}} \frac{1}{|\text{cb}|} \sum_t L_t^{\text{rprobe}} + \lambda_{\text{floor}} \text{ReLU}(\tau - \alpha_{\text{mem}})^2$ 
11: Backprop through  $\bar{L}$  into memres parameters; AdamW step.

```

and the writer stays uninformative because the router sends no gradient back. The recipe uses $\lambda_{\text{floor}} = 0.5$, $\tau = 0.10$.

Frozen backbone. Joint training of the backbone collapses the chain-specific evidence lift to zero on closed-set persona corpora because the unfrozen LM head learns the answer marginal directly. We freeze the backbone for all reported headline results. This costs us nothing on real conversational data: LongMemEval callback questions and answers are diverse enough that there is no template prior the LM head could learn to bypass memory.

3.2 The full training step

3.3 Reference recipe

The canonical **v24a** recipe (used for the headline result of §6) is:

```

--preset qwen3-0.6b-large      --memres_mode attention_parity
--router_recent_bias_init 4    --router_mem_bias_init 0
--memres_extract_source hidden_14  --memres_extract_input_norm
--memres_extraction_depth 4
--memres_writer_kind slot_attention --memres_slot_attention_iters 3
--memres_queries_init orthogonal  --memres_slot_positional
--memres_judge_qk_layernorm
--memres_readout_depth 4
--readout_probe_enabled --readout_probe_loss_weight 0.5
--readout_probe_warmup_steps 200
--alpha_mem_floor_aux_weight 0.5    --alpha_mem_floor_target 0.10
--freeze_backbone --lr 1e-4 --lr_backbone 0
--steps 1000 --warmup 200 --window_k 3
--callback_loss_weight 5.0 --score_tail_frac 1.0
--mask_evidence_session_loss
--kill_on_memory_collapse --kill_on_memory_collapse_min_step 200
--save_best_metric evidence_lift    --train_chains <longmemeval-train>

```

The backbone is Qwen3-0.6B [15] held frozen; only the memres parameters and the readout-probe head are trained.

4 Honest Evaluation Methodology

The evaluation choices are not optional. Across the campaign we encountered multiple checkpoints whose interpretation flipped between “state-of-the-project positive” and “zero or negative” depending on which evaluator was run. Our eventual practice was to require, for any cited memory result, the full battery of memory-on / memory-off / memory-shuffled / evidence-redacted measurements on a callback-aware window, plus an empirical-prior audit on the answer distribution.

4.1 Callback-aware standalone evaluation

Let S_c be the callback session of chain c and A_c its annotated answer-token mask. For a memory construction r we define the *callback cross-entropy*

$$\text{CE}_r^{\text{cb}}(c) = \frac{1}{|A_c|} \sum_{i \in A_c} -\log p_\theta(x_i \mid x_{<i}, M_{c,r}(c)).$$

The reported deltas are

$$\Delta_{\text{nm}-\text{m}}^{\text{cb}} = \text{CE}_{\text{nomem}}^{\text{cb}} - \text{CE}_{\text{mem}}^{\text{cb}}, \quad \Delta_{\text{sh}-\text{m}}^{\text{cb}} = \text{CE}_{\text{shuffle}}^{\text{cb}} - \text{CE}_{\text{mem}}^{\text{cb}}, \quad (8)$$

$$\Delta_{\text{evidence}} = \Delta_{\text{nm}-\text{m}}^{\text{cb}} - (\text{CE}_{\text{nomem, floor}}^{\text{cb}} - \text{CE}_{\text{mem, floor}}^{\text{cb}}). \quad (9)$$

$\Delta_{\text{nm}-\text{m}}^{\text{cb}}$ is the aggregate memory help on the actual answer tokens. $\Delta_{\text{sh}-\text{m}}^{\text{cb}}$ replaces $M_c(c)$ with $M_c(c')$ for a random different chain c' and tests *chain-specificity*: a positive $\Delta_{\text{nm}-\text{m}}^{\text{cb}}$ accompanied by an equally large $\Delta_{\text{sh}-\text{m}}^{\text{cb}}$ is just “memory adds a generic prior,” not recall. Δ_{evidence} is the chain-specific gain after subtracting the evidence-redacted floor: it asks whether the memory channel helps *more* when it is built from sessions that contain the answer than when it is built from sessions that do not.

Why this is necessary. On v14k (D4v2 corpus, frozen 0.6B), the standard chain evaluator reports $\Delta_{\text{nm}-\text{m}} = -0.085$ on the same checkpoint where the callback-aware evaluator reports $\Delta_{\text{nm}-\text{m}}^{\text{cb}} = +1.44$. The standard evaluator dilutes a localised callback effect across ~ 4 sessions of filler-and-evidence, driving its signal-to-noise below one. The callback-aware evaluator is the canonical evaluator in this paper.

4.2 Evidence-redacted memory floor

A positive $\Delta_{\text{nm}-\text{m}}^{\text{cb}}$ alone is not yet a memory result. On v14k, 1.37 of the 1.44 nats survive when M_c is built *without* the answer (replacing each evidence session with a distractor session). That 95% floor means the apparent benefit is mostly an evidence-blind “memory is on” prior and only $\Delta_{\text{evidence}} = +0.071$ nats is a chain-specific recall benefit.

The redaction operation is itself a possible source of error: an “evidence-redacted” run that secretly leaks the answer through some subtle path will inflate the floor and deflate Δ_{evidence} . We audit the redaction by comparing five variants – match (the standard in-distribution redaction), default-distractor pool, cross-chain non-evidence sessions, skip-evidence, and zeroed-token evidence. On v15a/best the spread across the four redactions is 0.038 nats, inside the pre-declared 0.05-nat band, and the default floor is slightly *worse* than cross/zero – the opposite direction of a leak. On v15b/best the spread is 0.0066 nats. A zero Δ_{evidence} on either is therefore not a redaction artifact.

4.3 Empirical-prior and base-CE audits

A final shortcut is the answer distribution itself. The D4v2 corpus templated callback questions like “Quick question, what was my favorite COLOR again?” / “Your favorite COLOR is RED.”

Table 1: Six failure modes that can confound a recurrent-memory result. “Diagnostic” is the test we use to detect each; “Default” is the corresponding methodological choice in this paper.

Name	Symptom	Diagnostic	Default
Token-averaged dilution	Standard chain eval reports $\Delta_{\text{nm-m}} \approx 0$ although callback tokens improve.	Callback-token-only CE; compare token counts.	Use callback-aware evaluator.
Joint-training invisibility	Unfrozen backbone learns the answer prior directly; Δ_{evidence} collapses to ~ 0 .	Freeze-vs-joint ablation; compare CE_{mem} to empirical prior.	Frozen backbone for headline numbers.
Category-cue leak	“Your favorite COLOR is __” reduces the answer to a 32-way closed set.	Fit empirical $P(\text{item} \mid \text{category})$; compare to CE_{mem} .	Use corpora without templated category cues.
Window leakage	Evidence enters the LM attention window directly.	Count evidence positions in the scored window.	Smaller k or forced evidence gap.
Redaction uncertainty	Zero Δ_{evidence} could be a leaky redaction rather than a real null.	Five-variant redaction spread within 0.05 nats.	Audit redaction explicitly; report spread.
Writer-objective failure	Removing the prior leaves $\Delta_{\text{evidence}} \approx 0$; LM-NLL alone does not bind.	D5 random-code stress test; readout-probe ablation.	Readout-probe auxiliary at $\lambda_{\text{rprobe}} = 0.5$.

Given the category, the answer is in a closed set of ~ 32 items. An empirical Laplace-smoothed prior $P(\text{item} \mid \text{category})$ fitted on the 5000-chain training set scores the 128-validation callback tokens at 2.91 nats. The joint-trained 0.6B run reaches 2.97 nats – within 0.06 of the prior. The unfrozen LM head has learned the template, not retrieval.

We therefore make three numbers part of the audit for any new corpus: (i) the empirical-prior CE, (ii) the bare-backbone base CE (no memory, no fine-tuning), and (iii) the cued-uniform CE (uniform over the closed set if the category cue narrows it). On D4v2 these are 2.91, 7.69 (Qwen3-0.6B base) / 8.76 (Qwen3-1.7B base), and 3.47 respectively. LongMemEval, by contrast, has no closed-set template – the prior CE on its callbacks is essentially the marginal language model entropy on diverse, free-form answers – which is one of the reasons the v24a result is interpretable.

4.4 Window-leakage counts

The chain trainer scores a window of k recent sessions; if any evidence session of the callback chain falls inside that window, the “memory” result is contaminated by direct attention from the LM head into the evidence. On D4v2 with the v15-default $k = 4$, 58.5% of training chains and 59.6% of validation chains had at least one evidence session in the callback window. At $k = 3$, 41.4% / 43.4%. Only $k = 1$ eliminates direct evidence visibility. We make the per-corpus, per- k leakage fraction a required figure in any future memory-architecture report.

5 Six Failure Modes

We catalogue the six failure modes that can make a recurrent-memory result misleading or that hide a real memory effect, with the diagnostic that exposes each.

Together, these define five distinct *pathways* from M_c or context to the LM head, only one of which is a chain-specific recurrent-memory result. The intended path is $C_t \rightarrow \text{writer} \rightarrow M_c \rightarrow m^t \rightarrow \text{LM}$. A content-blind memory prior, a backbone template prior, an in-window evidence

Table 2: **Headline.** Callback-aware in-domain evaluation on LongMemEval validation ($n=50$). Frozen Qwen3-0.6B + memres, identical recipe across rows; only the training corpus varies. LME = LongMemEval (11.2 M tok, 100% callback-supervised); LME+MSC = the LME+MSC+TV merge (63.8 M tok, 7% callback-supervised); synthd5 = templated random-code synthetic corpus (cross-domain reference). Δ_{sh-m}^{cb} near zero rules out a generic “memory adds context” confound. $\Delta_{nm-m}^{cb} = +0.34$ nats on v24a is the first non-trivial chain-specific leak-free end-to-end memory result in our 24-version campaign.

Run	Trained on	CE_{mem}	CE_{nomem}	Δ_{nm-m}^{cb}	Δ_{sh-m}^{cb}
v24a (ours)	LME	2.94	3.28	+0.3445	−0.0025
v24c	LME+MSC	5.84	5.45	−0.3966	−0.0138
v21c (ref)	synthd5	5.11	4.67	−0.4344	−0.0249

path, and an evaluation-method artifact are the four non-memory pathways the methodology of §4 is designed to rule out before a positive number can be cited.

6 Headline Result: LongMemEval

The recipe of §3 was applied verbatim to three training corpora, with all other hyperparameters held identical and all evaluation done on the LongMemEval validation chains ($n=50$). We report the callback-aware CE_{mem} , CE_{nomem} , $CE_{shuffle}$, Δ_{nm-m}^{cb} , and Δ_{sh-m}^{cb} in Table 2.

Why v24a worked. LongMemEval train consists of 450 chains, all with callback annotations. The mean chain length is 48.6 sessions, real-content density is 99.9% (511 real tokens per 512-token session), and the content is naturalistic multi-topic conversation. The writer therefore sees, on every training chain: (i) diverse, semantically rich content with ample representational headroom for the 0.6B backbone, evidenced by the training NLL dropping to 2.10 nats versus ~ 3.5 on synthd5; (ii) chain-specific readout-probe gradient on every example, pushing the writer toward chain-distinguishable M_c ; and (iii) genuinely long-horizon recall, since 48-session chains require compressing facts from up to ~ 47 sessions earlier (only the last 3 are in the trainer’s $k=3$ window).

Magnitude. A $\Delta_{nm-m}^{cb} = +0.34$ nat reduction on diverse free-form callback answers corresponds to a memory-on log likelihood ratio of $e^{0.34} \approx 1.40\times$ over the memory-off baseline, on tokens that the bare backbone has never seen the chain context for. The shuffle baseline at -0.003 nat shows the gain comes specifically from the matching chain’s evidence content rather than from any memory-on prior shift.

Per-chain distribution. Of the 50 scored validation chains, 40 have positive Δ_{nm-m}^{cb} (memory helps) and 10 have negative Δ_{nm-m}^{cb} (memory hurts), with the per-chain Δ_{nm-m}^{cb} ranging from $+3.00$ nats on a chain about a specific date narrowing the answer to a few tokens, down to -0.21 nats on a chain where the model’s no-memory marginal happens to land on the right answer. The shuffle distribution is much tighter, concentrated near zero with std ~ 0.05 nats, which is what the $\Delta_{sh-m}^{cb} = -0.003$ aggregate reflects.

7 Ablations and Negative Results

The headline result is preceded by a long sequence of negative and ambiguous results. We summarise the most informative ones; each isolates a distinct lever.

7.1 Corpus density: LME vs. LME+MSC merged (v24c)

v24c uses the same recipe but trains on a merged corpus comprising LongMemEval (450 chains) plus MSC (4000 chains) and ~ 1900 TV-script chains, none of the latter with callback annotations. Only 7% of training chains therefore receive readout-probe supervision; the other 93% contribute only LM-NLL, which teaches the writer to compress generic conversational content but supplies no chain-specific gradient channel.

End-to-end on lme_val: $\Delta_{\text{nm-m}}^{\text{cb}} = -0.40$. Memory *hurts*. The v24c writer compresses some mixture of LME-callback-relevant and MSC/TV-generic features that is actively worse than no memory for the LME callback. The implication for any future scaling is that the training corpus must have callback annotations on the bulk of chains; auxiliary corpora can only contribute to LM-NLL if synthetic callbacks are constructed for them.

7.2 Synthetic random-code corpora and seed instability (v21c, v23)

v21c was the strongest cell on the synthd5 random-code corpus, giving $\Delta_{\text{evidence}} = +0.024$ at end of training (on synthd5 validation, not LongMemEval). A multi-seed reproduction (v23a/b/c at seeds 1, 2, 7; the original is seed 42) showed the result was a single lucky seed: two of three new seeds collapsed into the v13-era uniform-fixed-point writer and were killed by the `--kill_on_memory_collapse` watchdog, and the surviving new seed ended at $\Delta_{\text{evidence}} = -0.0006$. Table 3 gives the four readings.

Table 3: v21c-recipe multi-seed reproducibility on synthd5 (**not** on LongMemEval). Two of four seeds collapse; the +0.024 figure is a single lucky seed, not a recipe.

	seed	end	last pair/self	end Δ_{evidence}	verdict
1	KILLED step 400		0.006	—	uniform collapse
2	KILLED step 600		0.006	—	uniform collapse
7	OK final 1000		0.032	-0.0006	no benefit
42	OK final 1000		0.075	+0.0241	lucky seed

The mechanism (Architectural Prior 5, in our internal terminology): LM-NLL on synthd5’s templated random-code answers is too permutation-symmetric to penalise uniform M_c . The strong α_{mem} -floor opens the depth router by step ~ 200 , which amplifies the LM-NLL gradient on the writer; the readout-probe gradient at $\lambda_{\text{rprobe}} = 0.5$ is then on a knife edge between specialising the writer and being overwhelmed.

The same recipe on LongMemEval (v24a) does not collapse. The naturalistic-content readout probe sees diverse answers that *cannot* be solved by a uniform M_c , so the probe gradient is permutation-asymmetric in a way the synthd5 probe gradient is not. The recipe is therefore stable on real conversational data and unstable on templated random-code data, which is the opposite of the folklore intuition that synthetic data is easier.

7.3 Joint training invisibility on closed-set persona (v15)

On the D4v2 closed-set persona corpus the joint-trained 0.6B checkpoint v15b reaches $\text{CE}_{\text{mem}} = 2.97$ nats, within 0.06 nats of the empirical category prior at 2.91. The unfrozen LM head has learned the template directly. The same architecture frozen-backbone (v14k/v15a) gives $\Delta_{\text{nm-m}}^{\text{cb}} = +1.44$ on the same checkpoint with $\Delta_{\text{evidence}} = +0.07$ – a real, but small, chain-specific gain. The lesson is methodological: report the backbone-only prior alongside any joint-training memory number.

7.4 Read-side capacity probe (§5 / TTT-on- M_c)

A complementary diagnostic asks whether the writer encoded the binding even if the readout/depth-router cannot use it end-to-end. The probe freezes the model and optimises only the readout’s W_Q^r, W_K^r, W_V^r on the callback tokens of a held-out chain set, then measures the chain-specific CE reduction. v20a (probe weight 0.3, depth=4) is the project’s best §5 reading at +0.120, while v21c (probe 0.5) – the best end-to-end on synthd5 – has §5 of −0.132. The two metrics are anti-correlated in this regime: high probe weight tightly fits the readout to the trained writer’s specific M_c distribution (good for end-to-end with that writer; bad for TTT-on- M_c which moves the writer); low probe weight keeps the readout flexible enough to decode any reasonable M_c . We treat end-to-end Δ_{nm-m}^{cb} as the primary metric; §5 is a useful complementary probe of whether *the architecture can* decode chain-specific M_c when given an oracle writer.

8 Related Work

Recurrent memory in Transformers. Transformer-XL [4] caches past hidden states within a window. Compressive Transformer [16] adds a coarse memory bank with hand-designed compression operators. Recurrent Memory Transformer [2] prepends a small bank of memory tokens to the input sequence and trains them recurrently across segments; its narrow channel and prepend-to-sequence design recreate the lost-in-the-middle pathology at long horizons. Block-Recurrent Transformer [8] introduces a learned gate to merge recurrent state with per-block hidden, but the gate is brittle to optimise. Memorising Transformers [19] retrieve from a key-value external memory but do not propagate gradient into the memory itself. Associative RMT [17] adds an explicit associative write head; AutoCompressors [3] compress context into soft prompts. Memory Residuals differ from all of these by reading off-sequence through a depth-wise router rather than prepending to or modifying the input sequence, and by separating extraction (Eq. 2) from competition (Eq. 3) so that the writer and the forgetting defence have distinct parameters.

Slot attention and object-centric latents. The slot-axis softmax of Eq. 3 is the slot-attention primitive of Locatello et al. [13] repurposed as the Stage-2 judge. The same primitive has been used in object-centric vision and in cross-modal segmentation; we use it here for its specific property of breaking the permutation-symmetric uniform fixed point.

Block attention residuals. Du et al. [5] propose a generalisation of the residual stream in which every Transformer sublayer receives a softmax-weighted mix over a pool of block-level sources rather than a uniform sum. This depth-wise routing pool is what we exploit to inject the memory readout off-sequence, registering m^t as a foundational source v_0 .

Long-context evaluation. Our findings align with a broad lesson from long-context evaluation: shortcuts are the default when the benchmark permits them. NoLiMa [14] removes literal overlap between query and evidence and shows ordinary long-context retrieval performance can be template-driven rather than robust memory use. RULER [7] argues vanilla needle-in-a-haystack is saturated and adds UUIDs, multiple needles, and aggregation to expose weaker retrieval behaviour. Lost-in-the-Middle [12] shows position effects even when context length is nominally large. LongBench [1] includes no-context checks to expose parametric priors; our callback-only and empirical-prior audits are the memory analogue. LongMemEval [20] separates indexing, retrieval, and reading, motivating our separation of write, readout, and decode diagnostics.

Honest evaluation, in general. The diagnostic methodology of this paper sits in the tradition of shortcut-learning critiques [6]: an architecture that *can* use a pathway is not the same as one

that *does* use it. Our six failure modes are concrete instances of that general principle for the recurrent-memory setting.

9 Discussion and Limitations

What v24a is and what it is not. v24a is a frozen-backbone 0.6B-parameter recurrent-memory module trained for 1000 steps on 11.2M tokens of LongMemEval, evaluated on 50 LongMemEval validation chains, that delivers $\Delta_{\text{nm-m}}^{\text{cb}} = +0.34$ nats of chain-specific callback-CE reduction with a near-zero shuffle baseline. It is not yet a state-of-the-art number on a public leaderboard, and it is not yet an end-to-end answer-accuracy result – the metric is callback-CE, not exact-match. The contribution is that this is the first cell in our 24-version campaign for which the diagnostic battery does not flip the conclusion, and the recipe transfers cleanly across training corpora as long as those corpora supply callback annotations on the bulk of chains.

Scale. The recipe is currently validated only at 0.6B. The natural next experiment is the same recipe on Qwen3-1.7B with the same training data; we expect a larger writer to compress more chain-specific information per slot, but earlier 1.7B runs (v15e on D4v2) showed a larger writer can also overfit non-evidence content into a generic “pre-route the callback” shortcut. The frozen-backbone constraint becomes more important, not less, at larger scale.

Real-world callback annotation. The recipe requires callback annotations on the bulk of training chains. LongMemEval is the largest public corpus we know of that supplies them. Scaling to larger corpora will require either synthetic callback construction on top of MSC/TV/PG-19 – which we have prototyped (Phase B in our internal scaling plan) but not yet validated – or a self-supervised proxy for the readout-probe loss that does not depend on explicit callback labels.

Joint training. Frozen-backbone is currently load-bearing because the unfrozen LM head learns the template prior on closed-set corpora. On LongMemEval, where there is no template prior, joint training may become safe; we have not yet run that experiment cleanly.

Open architectural levers. The slot-attention writer is one of several reasonable choices for the Stage-2 judge; we have not yet swept writer architectures with the v24a recipe. The depth- $D_R = 4$ readout refinement was added for read-side capacity but inflates $\|m^t\|/\|\text{embed}\|$ to $\sim 16\times$ at end of training, which suggests a magnitude bound or ReZero-init refinement may further stabilise late-training drift.

10 Conclusion

We presented **Memory Residuals**, a recurrent latent-memory primitive that augments a Transformer backbone with a fixed-size matrix M_c , updated through a two-stage QKV competition with a slot-axis softmax that breaks the symmetric uniform fixed point, and read back through a depth-wise attention-residual router that lets each layer choose between current-session computation and cross-session memory. Combined with the readout-probe auxiliary objective, an α_{mem} load-balance floor, a frozen backbone, and training on real callback-supervised conversational data, this yields $\Delta_{\text{nm-m}}^{\text{cb}} = +0.34$ nats of chain-specific callback-CE reduction on LongMemEval validation – the first non-trivial, chain-specific, leak-free end-to-end memory benefit in our campaign. The same architecture is unstable when the training corpus is templated random codes or when most training chains lack callback supervision, which we read as a corpus-density and writer-supervision result. We release the architecture, the recipe, the evaluation tooling, and the run ledger in the hope that the diagnostic methodology – callback-aware evaluation,

evidence-redacted floor, empirical-prior audit, window-leakage counts, and shuffle controls – becomes a baseline for any future recurrent-memory result.

Reproducibility Statement

All numbers in the main text are drawn from the per-chain callback-aware evaluation outputs and the run ledger of the project repository. The reference recipe is reproduced verbatim in §3.3; the architecture follows §2 and the training loop follows Algorithm 1. The canonical evaluator is the callback-aware standalone evaluator described in §4; the audit tools cover window-leakage counts (Appendix B), redaction-spread variance (Appendix C), and the empirical category-prior fit. Code, training scripts, evaluation tooling, and the v24a checkpoint will be released alongside the camera-ready version.

References

- [1] Bai, Y. *et al.* LongBench: A bilingual, multitask benchmark for long context understanding. arXiv:2308.14508, 2024.
- [2] Bulatov, A., Kuratov, Y., & Burtsev, M. S. Recurrent memory transformer. In *Advances in Neural Information Processing Systems*, 2022.
- [3] Chevalier, A., Wettig, A., Ajith, A., & Chen, D. Adapting language models to compress contexts. In *EMNLP*, 2023.
- [4] Dai, Z., Yang, Z., Yang, Y., Carbonell, J., Le, Q. V., & Salakhutdinov, R. Transformer-XL: Attentive language models beyond a fixed-length context. In *ACL*, 2019.
- [5] Du, Y. *et al.* (Kimi Team). Attention residuals. arXiv:2603.15031, 2026.
- [6] Geirhos, R., Jacobsen, J.-H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M., & Wichmann, F. A. Shortcut learning in deep neural networks. arXiv:2004.07780, 2020.
- [7] Hsieh, C.-P. *et al.* RULER: What’s the real context size of your long-context language models? arXiv:2404.06654, 2024.
- [8] Hutchins, D., Schlag, I., Wu, Y., Dyer, E., & Neyshabur, B. Block-recurrent transformers. In *Advances in Neural Information Processing Systems*, 2022.
- [9] Jaegle, A. *et al.* Perceiver IO: A general architecture for structured inputs and outputs. In *ICLR*, 2021.
- [10] Lewis, P. *et al.* Retrieval-augmented generation for knowledge-intensive NLP tasks. In *NeurIPS*, 2020.
- [11] Liu, Y. Memory residuals: Innate working memory via depth-wise attention. Position paper, 2025.
- [12] Liu, N. F. *et al.* Lost in the middle: How language models use long contexts. *TACL*, 12:157–173, 2024.
- [13] Locatello, F., Weissenborn, D., Unterthiner, T., Mahendran, A., Heigold, G., Uszkoreit, J., Dosovitskiy, A., & Kipf, T. Object-centric learning with slot attention. In *NeurIPS*, 2020.
- [14] Modarressi, A. *et al.* NoLiMa: Long-context evaluation beyond literal matching. arXiv:2502.05167, 2025.

- [15] Qwen Team. Qwen3 technical report. 2024.
- [16] Rae, J. W., Potapenko, A., Jayakumar, S. M., & Lillicrap, T. P. Compressive transformers for long-range sequence modelling. In *ICLR*, 2019.
- [17] Rodkin, I. *et al.* Associative recurrent memory transformer. ICML 2024 Next Generation of Sequence Modeling Architectures workshop, 2024.
- [18] Vaswani, A. *et al.* Attention is all you need. In *NeurIPS*, 2017.
- [19] Wu, Y., Rabe, M. N., Hutchins, D., & Szegedy, C. Memorising transformers. arXiv:2203.08913, 2022.
- [20] Yu, D. *et al.* LongMemEval: Benchmarking chat assistants on long-term interactive memory. arXiv:2410.10813, 2025.
- [21] Zhai, S. *et al.* Stabilizing Transformer training by preventing attention entropy collapse. In *ICML*, 2023.

A Per-corpus base CE and prior CE

Table 4: Base callback CE without memory or fine-tuning, on the callback-mask tokens of each corpus.

Backbone / baseline	D4v2	LongMemEval
Qwen3-0.6B base	7.69	3.28
Qwen3-1.7B base	8.76	—
Uniform over closed-set items (256 / open)	5.55	—
Cued uniform over 32-item category	3.47	n/a (no template)
Empirical $P(\text{item} \mid \text{cat})$	2.91	n/a (no template)

B Window leakage counts on D4v2

For D4v2, evidence positions are sampled from body positions $0, \dots, 8$ and the callback is at position 9. Audit A1 counts whether an evidence session falls in the callback scoring window of size k :

k	train chains leaking / 5000	val chains leaking / 500
4	2926 (58.5%)	298 (59.6%)
3	2068 (41.4%)	217 (43.4%)
2	1088 (21.8%)	114 (22.8%)
1	0	0

Only $k = 1$ eliminates direct evidence visibility.

C Redaction-spread audit

A2 evaluates five floor variants: original evidence (match), default distractor-pool redaction, cross-chain non-evidence redaction, skip-evidence, and zeroed-token evidence. On v15a/best, the redaction-style spread is 0.038 nats; on v15b/best it is 0.0066 nats. The default floor is slightly worse than cross/zero, which is the opposite direction of a leak.

D Per-chain headline data

The full 50-chain per-chain CE_{mem} , CE_{nomem} , CE_{shuffle} on LongMemEval validation under v24a will be released alongside the camera-ready version. The distribution of per-chain $\Delta_{\text{nm-m}}^{\text{cb}}$ has 40/50 chains positive, 10/50 negative, mean +0.345 nats, median +0.13 nats, max +3.00 nats, min -0.21 nats.